GROUP B) 4)

localhost:8888/notebooks/Desktop/Group%20C/B_4_Tips.ipynb

jupyter B_4_Tips (unsaved changes)

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```
In [1]: import pandas as pd
```

```
In [11]: df = pd.read_csv('tips_DV.csv')
```

```
In [12]: df
```

Out[12]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

```
In [13]: df.rename(columns = {'size':'peoples'}, inplace = True)
```

```
In [14]: df
```

Out[14]:

	total_bill	tip	sex	smoker	day	time	peoples
0	16.99	1.01	Female	No	Sun	Dinner	2

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3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

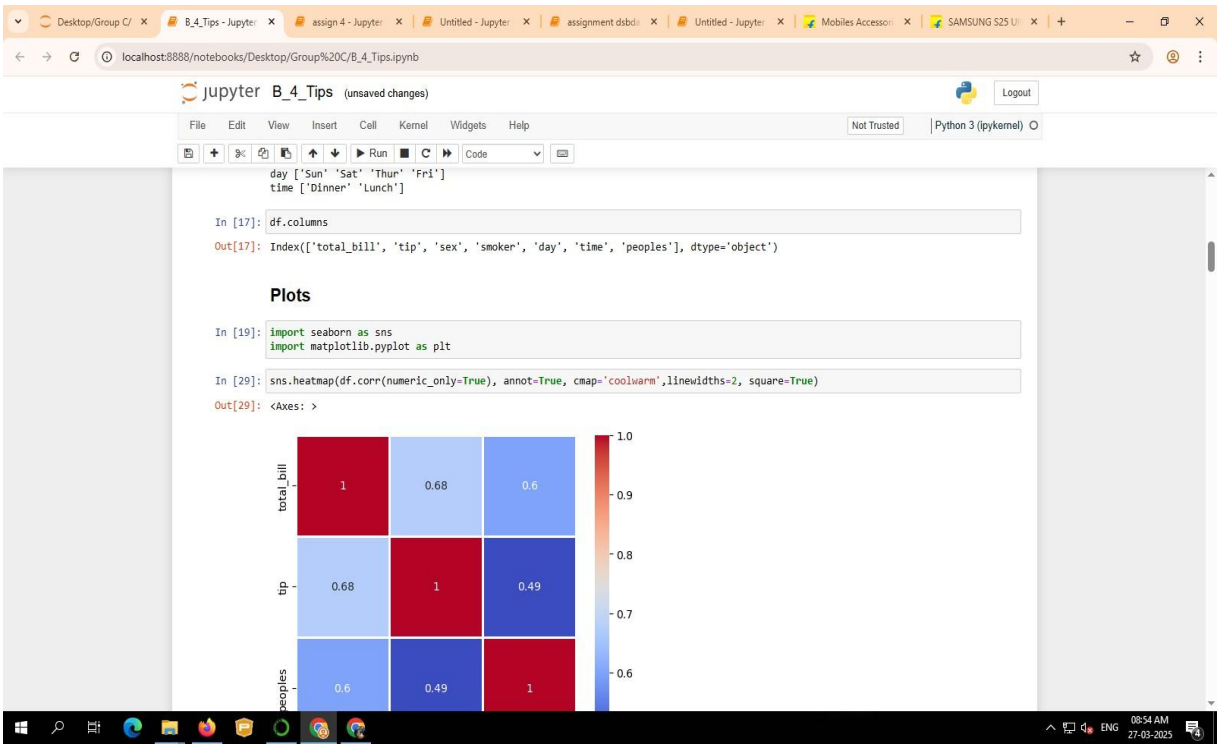
```
In [15]: df.isna().sum()
```

```
Out[15]: total_bill    0
         tip           0
         sex           0
         smoker        0
         day           0
         time          0
         peoples       0
         dtype: int64
```

```
In [16]: col_var = df[['peoples','sex', 'smoker', 'day', 'time']]

         for i in col_var:
             print(i, df[i].unique())
```

```
peoples [2 3 4 1 6 5]
sex ['Female' 'Male']
smoker ['No' 'Yes']
day ['Sun' 'Sat' 'Thur' 'Fri']
time ['Dinner' 'Lunch']
```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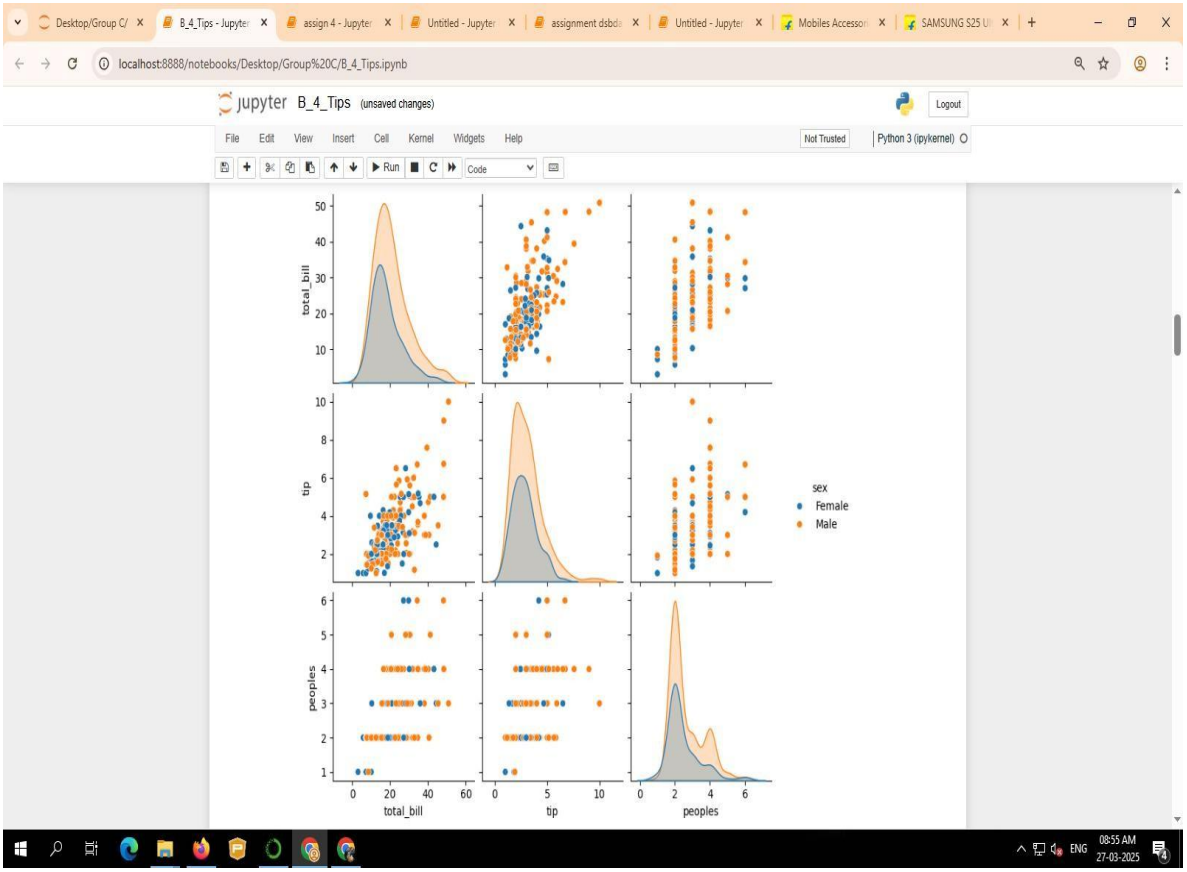
Not Trusted
Python 3 (ipykernel)
day ['Sun' 'Sat' 'Thur' 'Fri']
time ['Dinner' 'Lunch']
In [17]: df.columns
Out[17]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'peoples'], dtype='object')
Plots
In [19]: import seaborn as sns
import matplotlib.pyplot as plt
In [29]: sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm',linewidths=2, square=True)
Out[29]: <Axes: >
total_bill
tip
peoples
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
1.0
0.9
0.8
0.7
0.6


jupyter B_4_Tips (unsaved changes)
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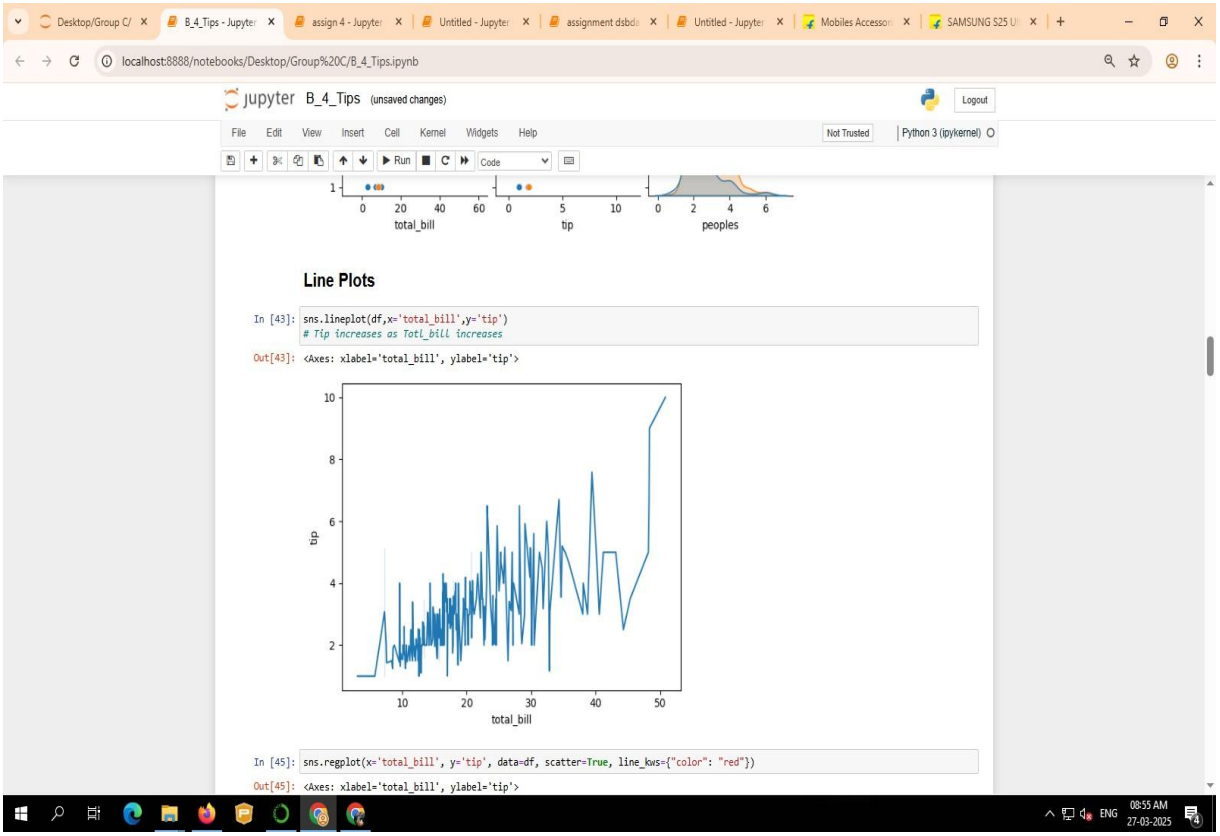
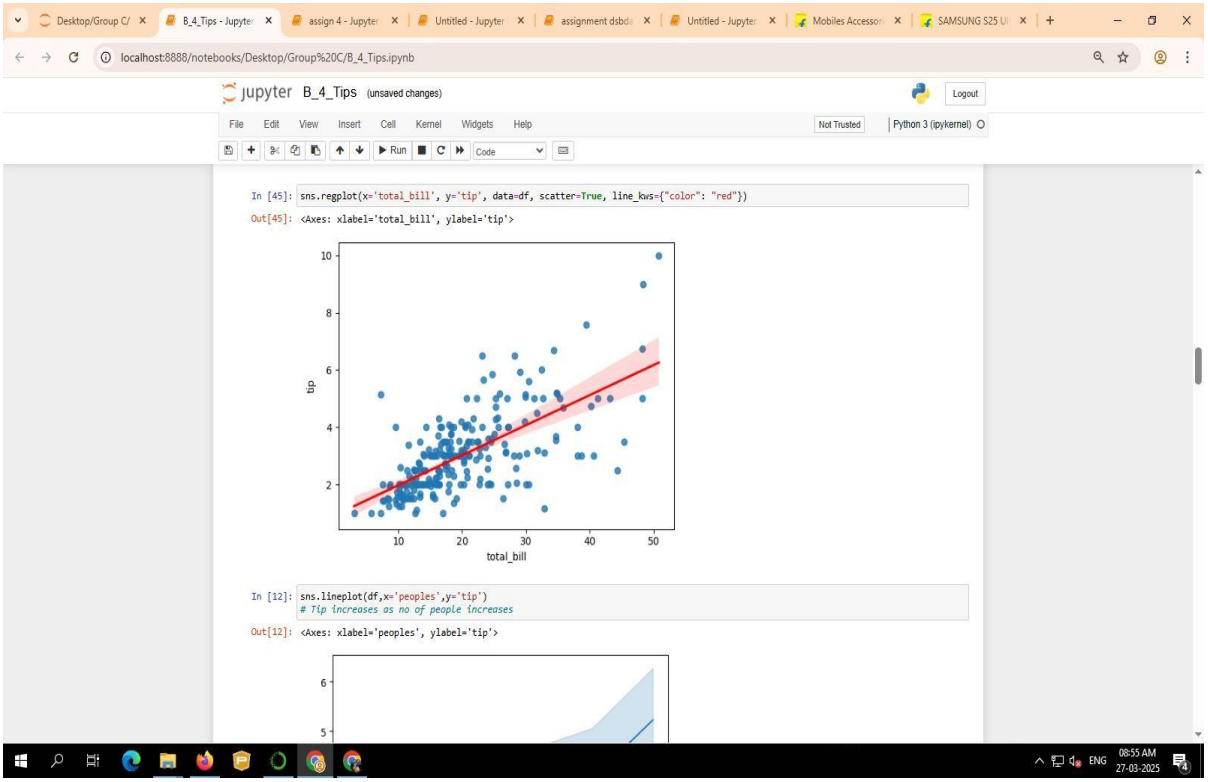
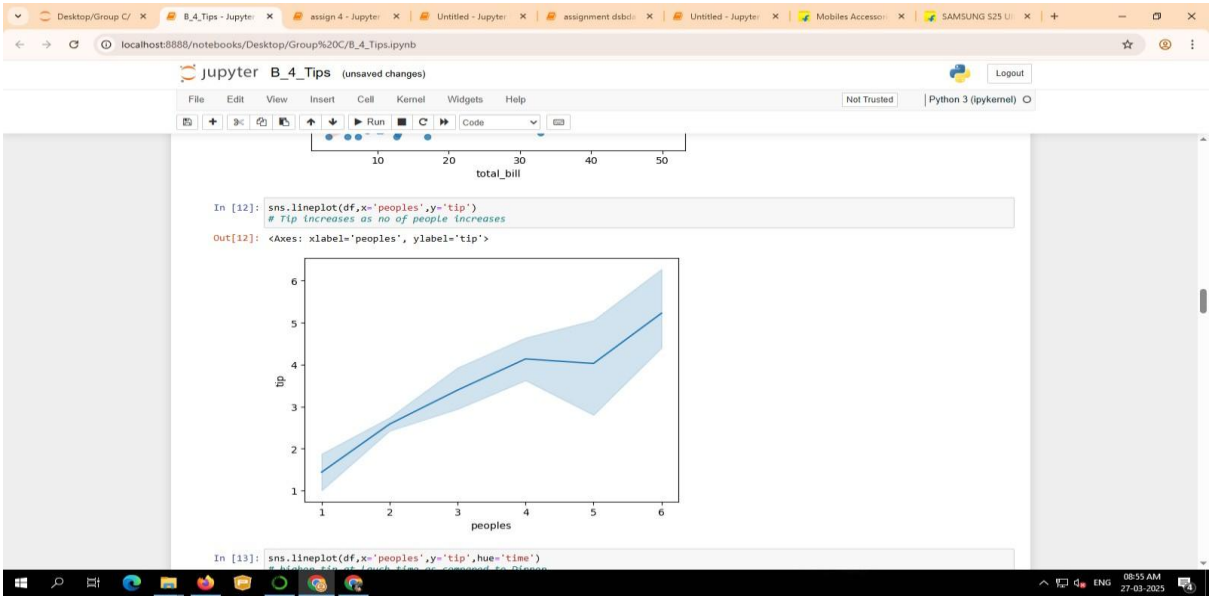
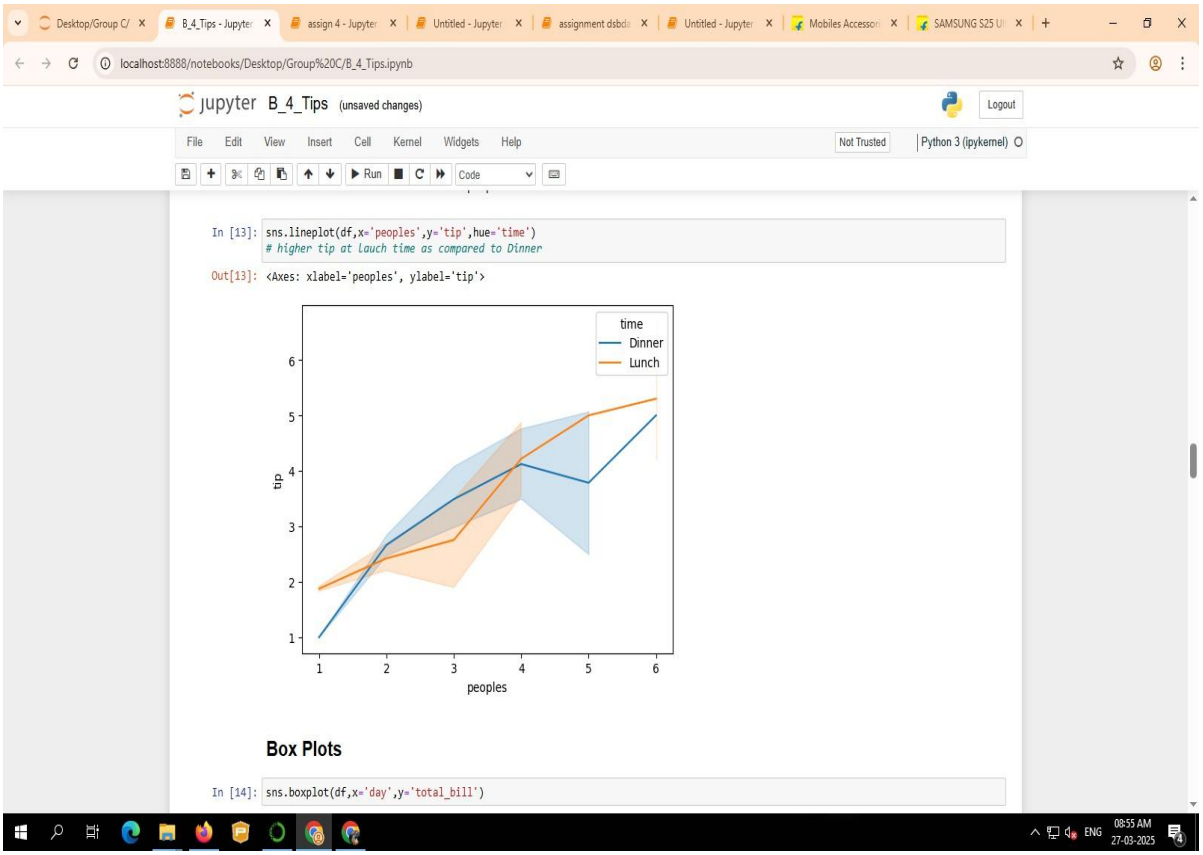
Not Trusted
Python 3 (ipykernel)
total_bill
tip
peoples
sex
Female
Male


jupyter B_4_Tips (unsaved changes)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
total_bill
tip
peoples
Line Plots
In [43]: sns.lineplot(df,x='total_bill',y='tip')
Tip increases as Totl_bill increases
Out[43]: <Axes: xlabel='total_bill', ylabel='tip'>
tip
total_bill
In [45]: sns.regplot(x='total_bill', y='tip', data=df, scatter=True, line_kws={"color": "red"})
Out[45]: <Axes: xlabel='total_bill', ylabel='tip'>


jupyter B_4_Tips (unsaved changes)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
In [45]: sns.regplot(x='total_bill', y='tip', data=df, scatter=True, line_kws={"color": "red"})
Out[45]: <Axes: xlabel='total_bill', ylabel='tip'>
tip
total_bill
In [12]: sns.lineplot(df,x='peoples',y='tip')
Tip increases as no of people increases
Out[12]: <Axes: xlabel='peoples', ylabel='tip'>


jupyter B_4_Tips (unsaved changes)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
total_bill
In [12]: sns.lineplot(df,x='peoples',y='tip')
Tip increases as no of people increases
Out[12]: <Axes: xlabel='peoples', ylabel='tip'>
tip
peoples
In [13]: sns.lineplot(df,x='peoples',y='tip',hue='time')


jupyter B_4_Tips (unsaved changes)
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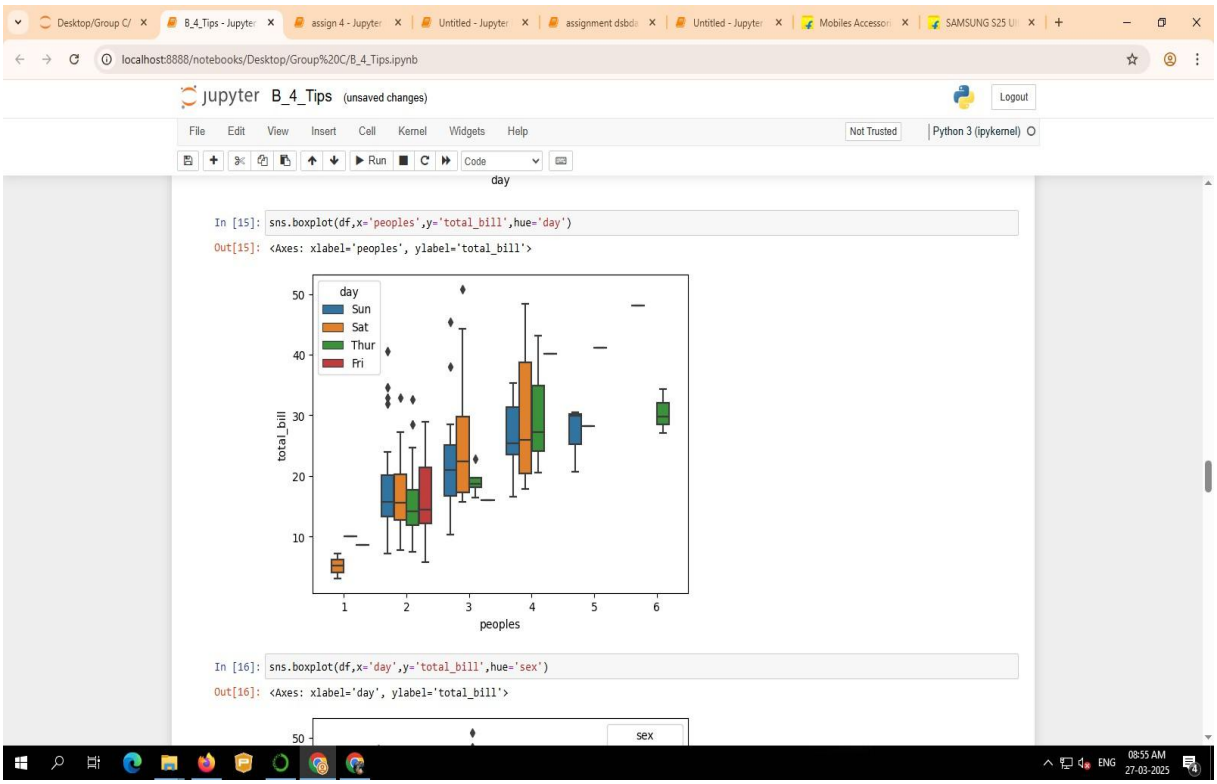
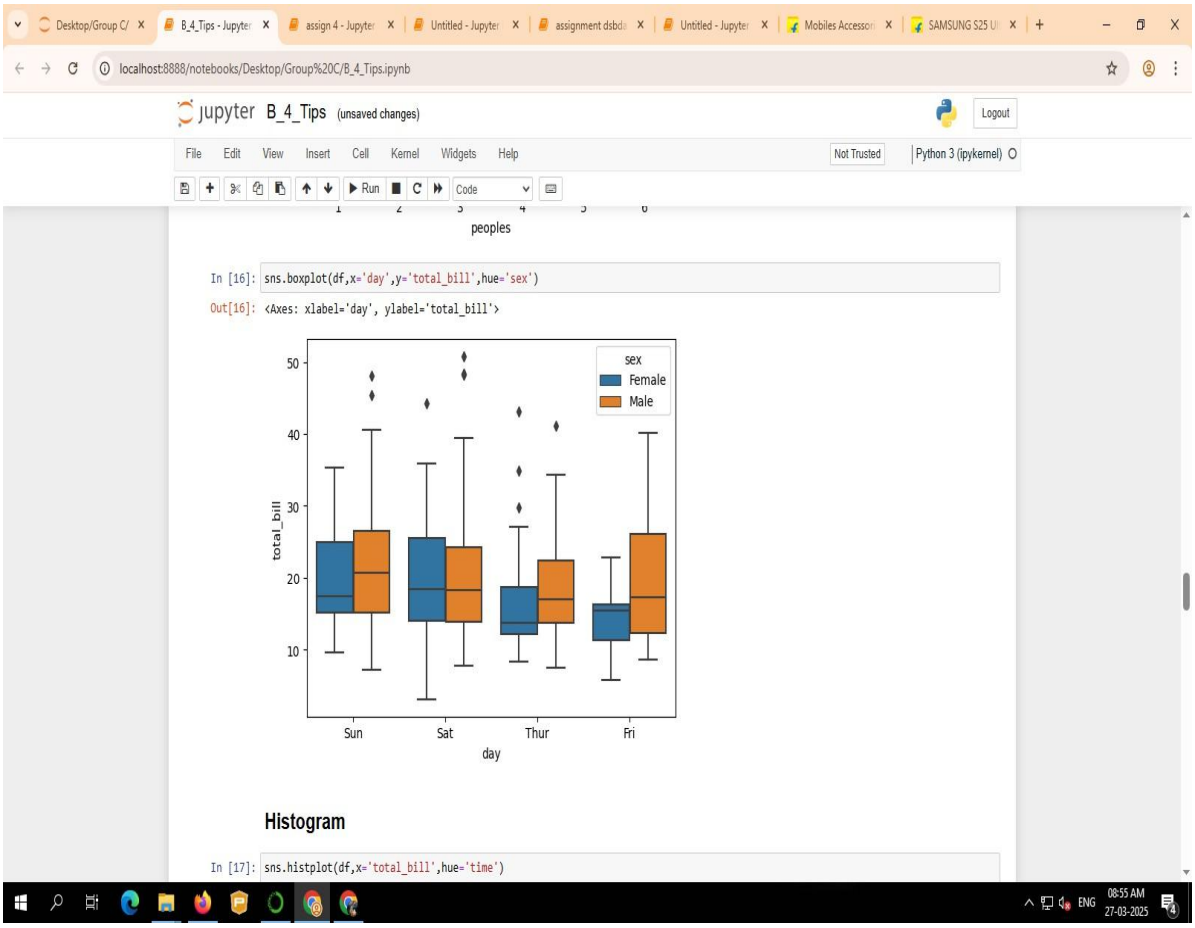
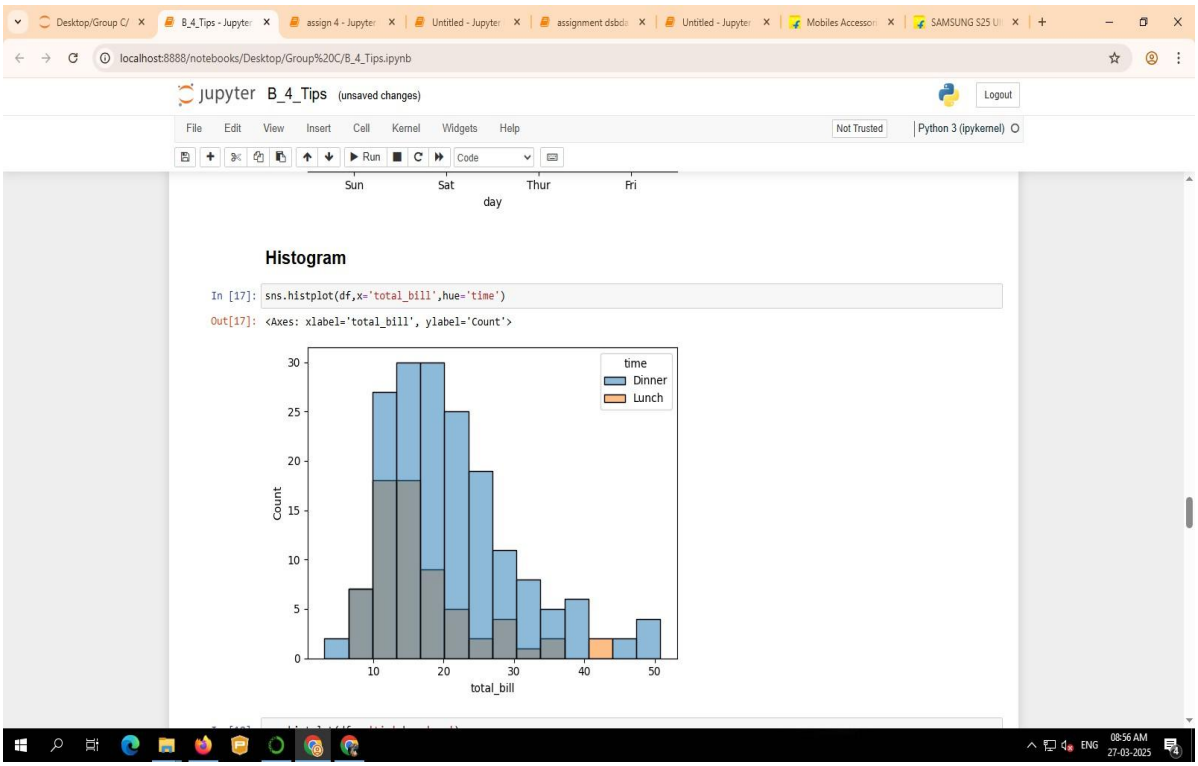
Not Trusted
Python 3 (ipykernel)
In [13]: sns.lineplot(df,x='peoples',y='tip',hue='time')
higher tip at lauch time as compared to Dinner
Out[13]: <Axes: xlabel='peoples', ylabel='tip'>
time
Dinner
Lunch
tip
peoples
Box Plots
In [14]: sns.boxplot(df,x='day',y='total_bill')

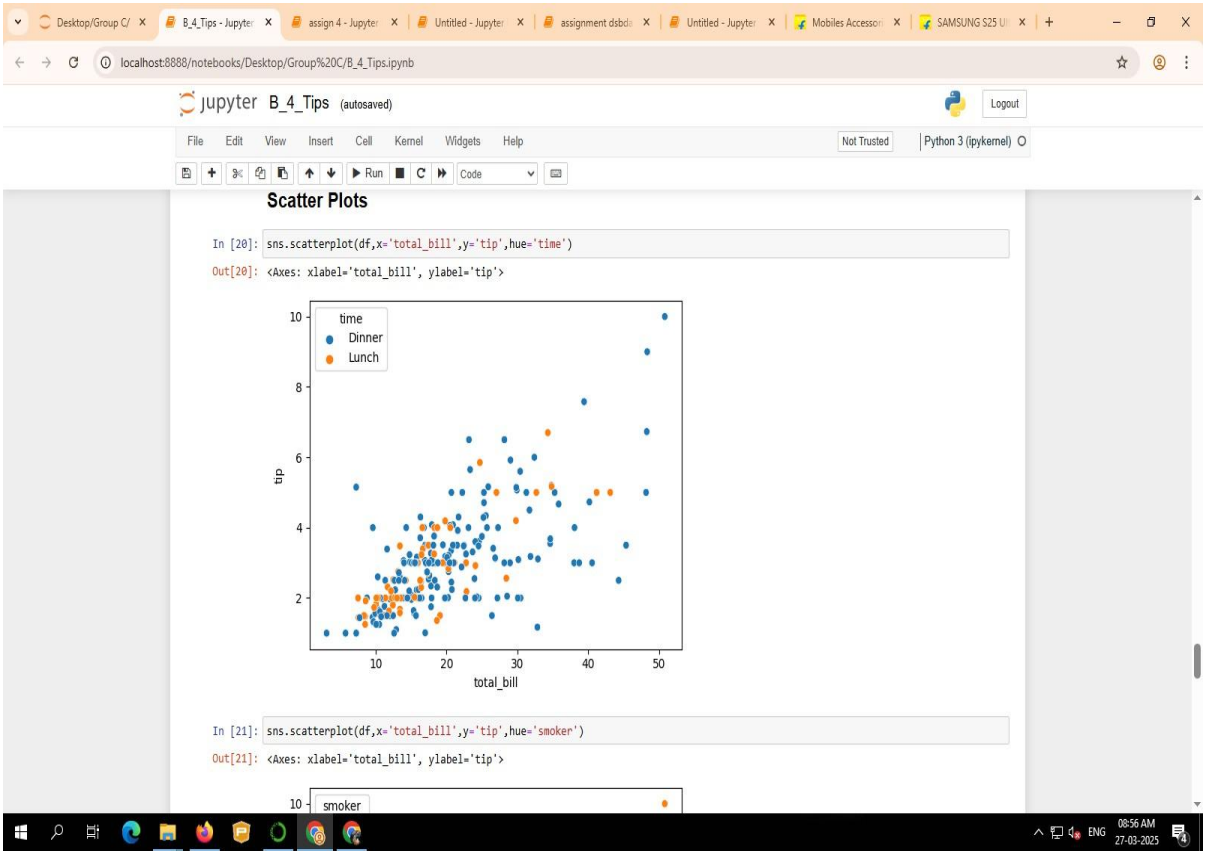




Histogram

```
In [17]: sns.histplot(df,x='total_bill',hue='time')
```


jupyter B_4_Tips (unsaved changes)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
Sun
Sat
Thur
Fri
day
Histogram
In [17]: sns.histplot(df,x='total_bill',hue='time')
Out[17]: <Axes: xlabel='total_bill', ylabel='Count'>
time
Dinner
Lunch
Count
total_bill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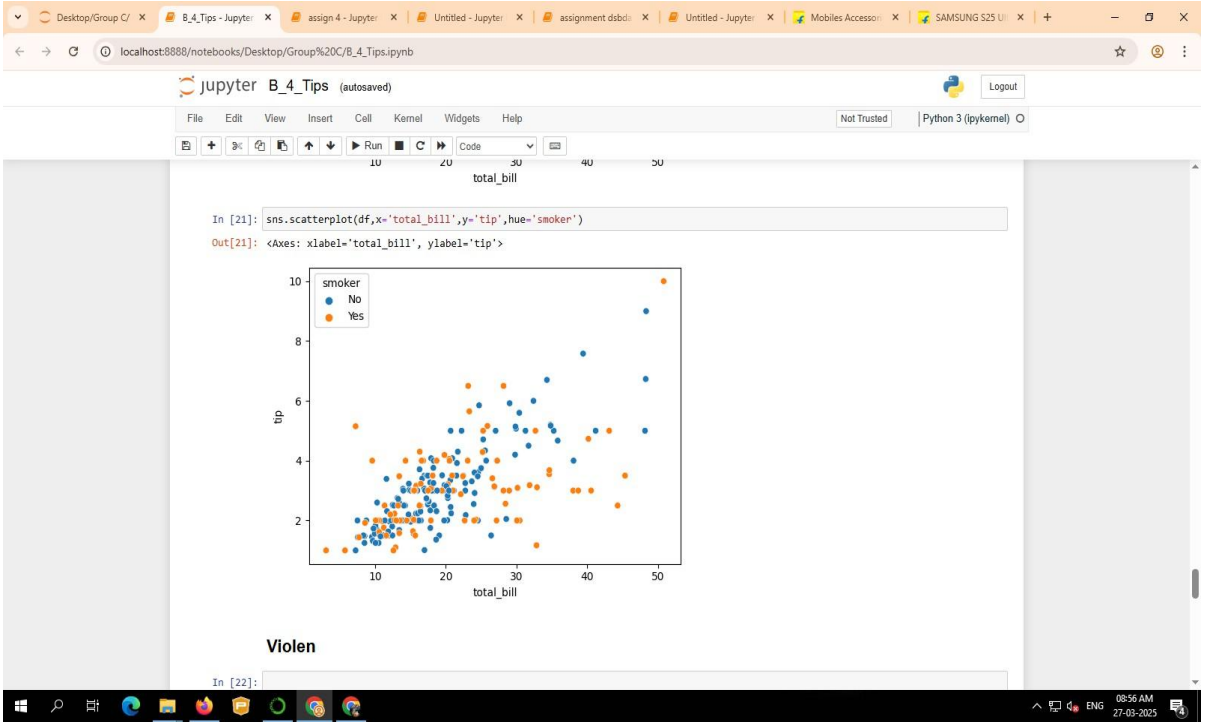
Python 3 (ipykernel)
Scatter Plots
In [20]: sns.scatterplot(df,x='total_bill',y='tip',hue='time')
Out[20]: <Axes: xlabel='total_bill', ylabel='tip'>
time
Dinner
Lunch
tip
total_bill
In [21]: sns.scatterplot(df,x='total_bill',y='tip',hue='smoker')
Out[21]: <Axes: xlabel='total_bill', ylabel='tip'>
smoker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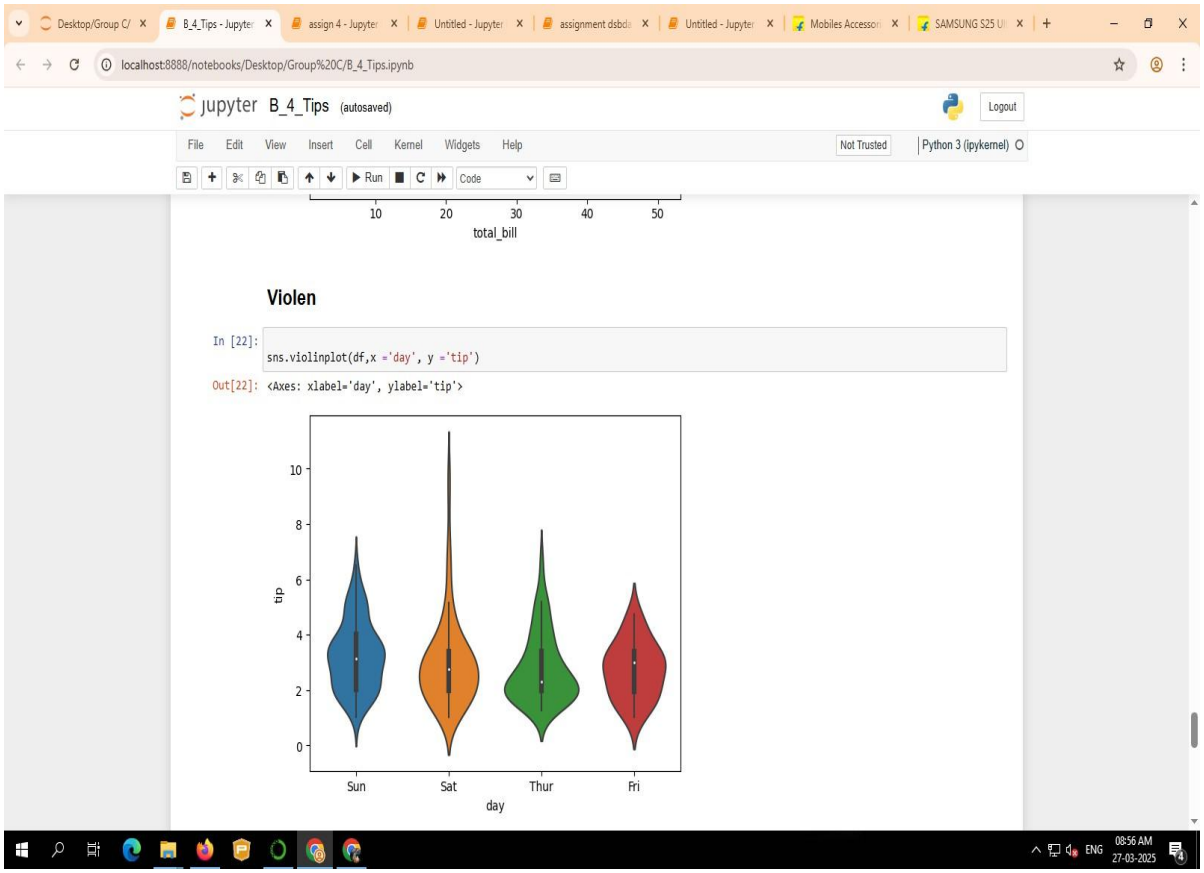
Not Trusted
Python 3 (ipykernel)
In [21]: sns.scatterplot(df,x='total_bill',y='tip',hue='smoker')
Out[21]: <Axes: xlabel='total_bill', ylabel='tip'>
smoker
No
Yes
tip
total_bill
Violen
In [22]: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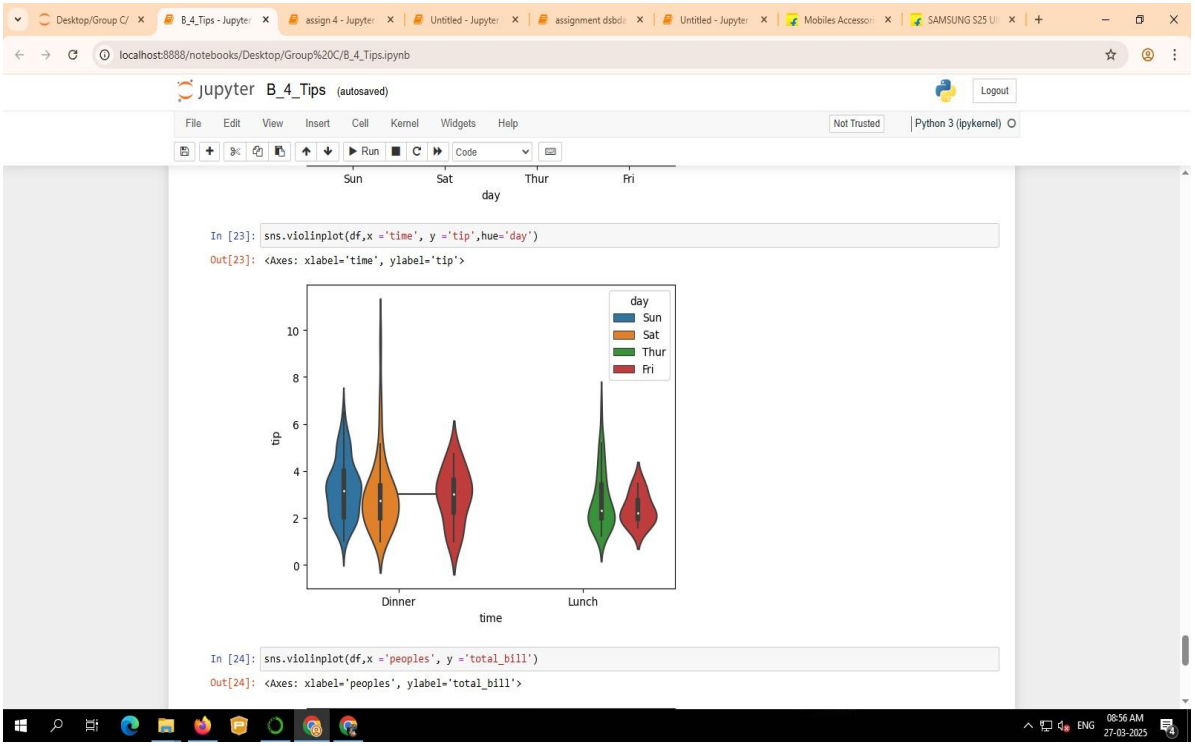
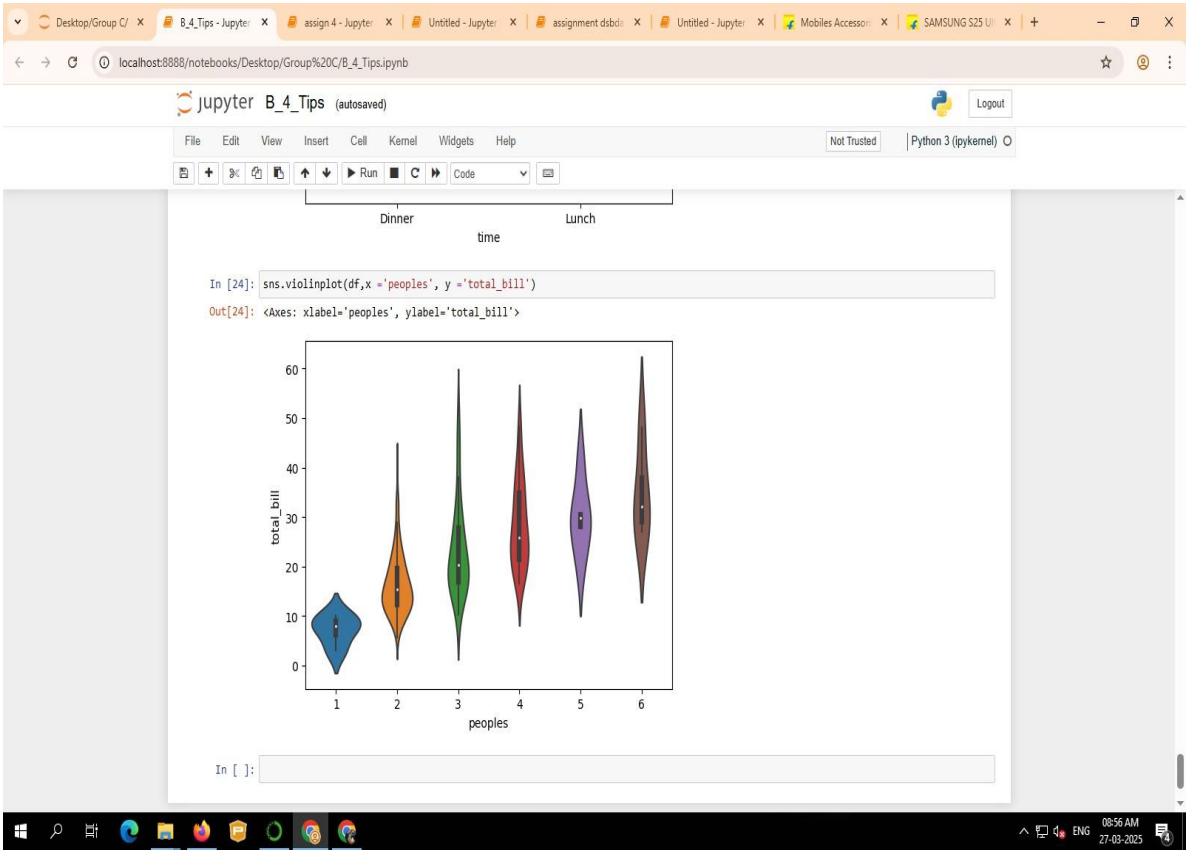
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
total_bill
Violen
In [22]:
sns.violinplot(df,x ='day', y ='tip')
Out[22]: <Axes: xlabel='day', ylabel='tip'>
tip
Sun
Sat
Thur
Fri
day


jupyter B_4_Tips (autosaved)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
Sun Sat Thur Fri
day
In [23]: sns.violinplot(df,x ='time', y ='tip',hue='day')
Out[23]: <Axes: xlabel='time', ylabel='tip'>
day
Sun
Sat
Thur
Fri
tip
Dinner
Lunch
time
In [24]: sns.violinplot(df,x ='peoples', y ='total_bill')
Out[24]: <Axes: xlabel='peoples', ylabel='total_bill'>


jupyter B_4_Tips (autosaved)
File Edit View Insert Cell Kernel Widgets Help
Not Trusted
Python 3 (ipykernel)
Dinner
Lunch
time
In [24]: sns.violinplot(df,x ='peoples', y ='total_bill')
Out[24]: <Axes: xlabel='peoples', ylabel='total_bill'>
total_bill
peoples
In [ ]: